# TLS_Pie Raspberry Pi Setup Guide

This guide collects the simplest setup path for the Raspberry Pi used by the TLS_Pie lidar project.

## 1. What you need

A Raspberry Pi (Pi 4 recommended)
A microSD card (16 GB or larger)
A power supply for the Pi
A monitor, keyboard, and mouse if you want to use the desktop UI
An Ethernet cable for the lidar connection
Internet access for the Pi

## 2. Download Raspberry Pi OS

Download Raspberry Pi OS from:
https://www.raspberrypi.com/software/operating-systems/

Recommended choice:
Raspberry Pi OS with desktop

## 3. Write the image to the microSD card

Use Raspberry Pi Imager:
https://www.raspberrypi.com/software/

Steps:
1. Open Raspberry Pi Imager
2. Choose Raspberry Pi OS
3. Choose your microSD card
4. Click Write

## 4. Boot the Pi

1. Insert the SD card into the Pi
2. Connect power
3. Wait for the Pi to boot
4. Log in to the desktop

## 5. Connect the Pi to the network

Connect the Pi to your network with Ethernet if possible
Make sure the Pi can reach the internet

## 6. Open a terminal

Open the terminal on the Pi.

## 7. Run the setup script

Make sure the project files are on the Pi at:
/home/lipi/TLS-Pie

Then run:

```
sudo bash /home/lipi/TLS-Pie/Raspberry\ Pie4/setup_tls_pie_pi.sh
```

This script will install the required packages and set up the auto-start entry.

## 8. Test the recording script manually

Run:

sudo /home/lipi/TLS-Pie/Raspberry\ Pie4/TLS-Pie/VLPrecord.sh eth0

If your interface is not eth0, check it with:

ip -br addr

Then use the correct name.

## 9. Reboot and verify

Reboot the Pi:

sudo reboot

After reboot, confirm:
the script starts automatically
the capture directory exists
a .pcap file is created

## 10. MicroView display behavior and wiring

MicroView now shows clear runtime states:
READY when idle and prepared to scan
SCANNING during active scan motion
REC once Pi confirms tcpdump recording started
PI TIMEOUT if no Pi recording acknowledgement is received
PI ABORTED with a reason code if Pi aborts

Add one Pi-to-MicroView return signal line through an unused level shifter channel:
Pi GPIO22 -> level shifter -> MicroView D4 (PISTATUS)

Abort cause codes shown on OLED:
1 START TIMEOUT
2 NO INTERFACE
3 LIDAR OFFLINE
4 TCPDUMP ERROR
5 EMPTY PCAP
6 INTERRUPTED
7 TOOL MISSING

## 11. Common issues

The lidar interface is not eth0
Missing packages
The Pi cannot reach the network
The Arduino and Pi do not share a common ground

## 12. Useful links

Raspberry Pi OS downloads: https://www.raspberrypi.com/software/operating-systems/
Raspberry Pi Imager: https://www.raspberrypi.com/software/
Raspberry Pi documentation: https://www.raspberrypi.com/documentation/